PROBLEM 5. VLE Data Reduction with NRTL

A. $\ln\gamma_1^{\infty} = \ln\gamma_1(x_1 \to 0, x_2 \to 1)$

From the given NRTL equation for $\ln\gamma_1$ we obtain by substituting $x_1 = 0$ and $x_2 = 1$

$$\ln\gamma_1^{\infty} = \tau_{21} + \tau_{12} G_{12} \Rightarrow \ln\gamma_1^{\infty} = \tau_{21} + \tau_{12}\exp(-0.3\,\tau_{12})$$

In similar manner

$\ln\gamma_2^{\infty} = \ln\gamma_2(x_2 \to 0, x_1 \to 1) \Rightarrow$

$$\ln\gamma_2^{\infty} = \tau_{12} + \tau_{21} G_{21} \Rightarrow \ln\gamma_2^{\infty} = \tau_{12} + \tau_{21}\exp(-0.3\,\tau_{21})$$

B. To determine whether the experimental data are thermodynamically consistent we need to estimate the integral

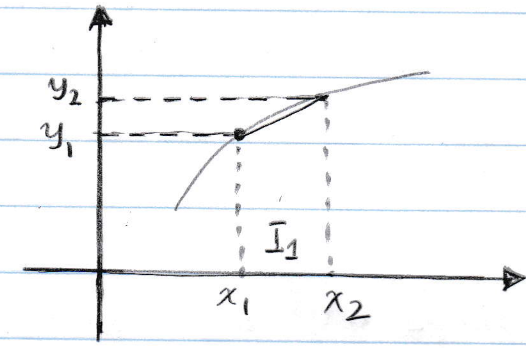
$$I = \int_0^1 \ln\left(\frac{\gamma_1}{\gamma_2}\right) dx_1$$

First we notice that $K_1 = \dfrac{y_1}{x_1} = \gamma_1 \dfrac{P_1^{sat}}{P} \Rightarrow$

$$\gamma_1 = \frac{y_1}{x_1}\frac{P}{P_1^{sat}} \quad \text{and also} \quad \gamma_2 = \frac{y_2}{x_2}\frac{P_2^{sat}}{P}$$

From these last two equations we can calculate the value of $\ln\gamma_1 - \ln\gamma_2 = \ln\left(\frac{\gamma_1}{\gamma_2}\right)$

The integral $\int_0^1 \ln\left(\frac{\gamma_1}{\gamma_2}\right) dx_1$ can be calculated using the trapezoid rule. Looking at the graph below the trapezoid rule is given by



$$I_1 = \frac{y_1 + y_2}{2}(x_2 - x_1)$$

In order to calculate the integral we need values of γ_1^{∞} and γ_2^{∞}. To find these values we perform a linear extrapolation from the two closest points

$$\gamma_1^{\infty} \cong 1.0555 + \frac{1.0555 - 1.0303}{0.0353 - 0.0700}(0 - 0.0353) \Rightarrow \gamma_1^{\infty} \approx 1.0811$$

$$\gamma_2^{\infty} = 1.2899 + \frac{1.2899 - 1.2246}{0.1383 - 0.2231}(0 - 0.1383) \Rightarrow \gamma_2^{\infty} \approx 1.3963$$

Better values of the liquid activity coefficients at infinite dilution can be obtain by consider quadratic approximations.

Now that we know the values of γ_1 and γ_2 for the entire range of $x_1 \in [0, 1]$ we can calculate the integral using the trapezoid rule. The numerical result as well the plot of $\ln(\gamma_1/\gamma_2)$ vs. x_1 are shown below:

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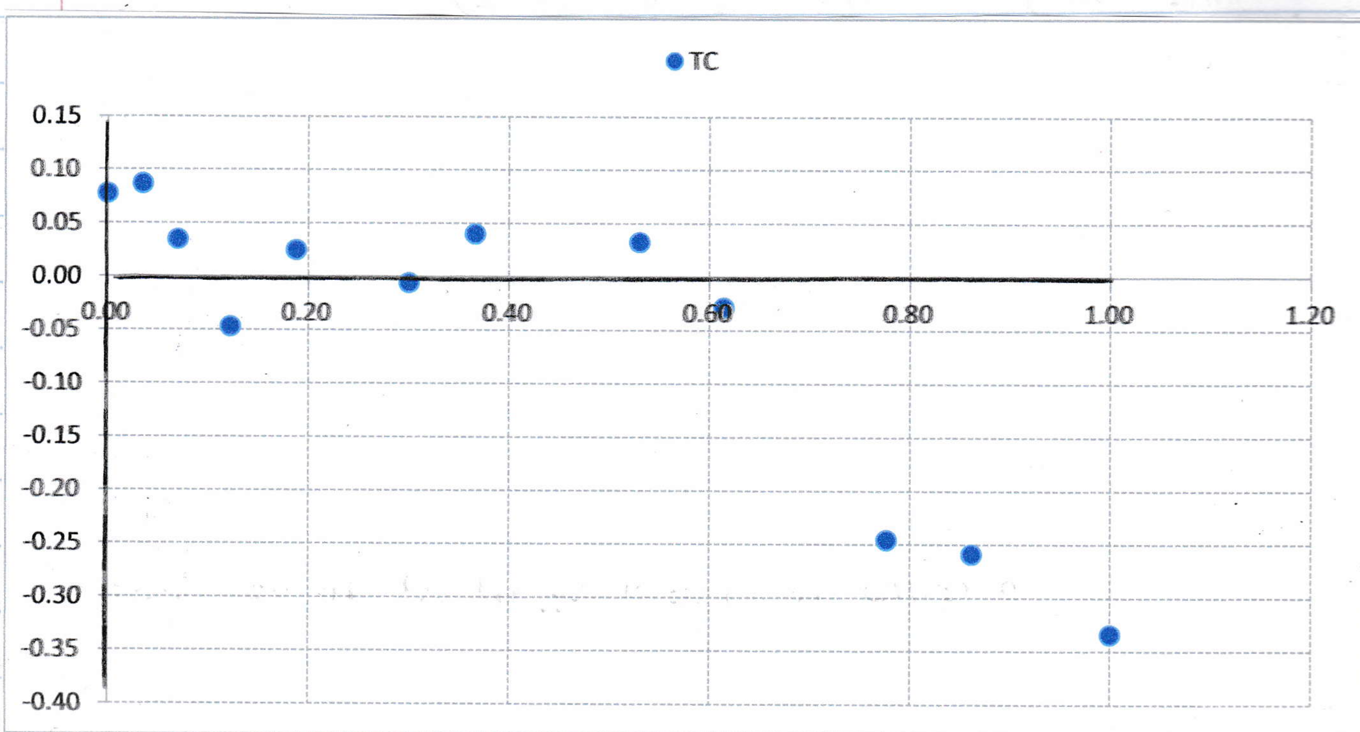
Plot of $\ln(\gamma_1/\gamma_2)$ vs x_1



Calculation of the integral under the curve

γ_1	γ_2	$\ln(\gamma_1) - \ln(\gamma_2)$	X_1	Section Integral
1.0811	1.0000	0.0780	0.0000	0.002912
1.0555	0.9676	0.0869	0.0353	0.002116
1.0303	0.9949	0.0350	0.0700	-0.000310
0.9625	1.0087	-0.0469	0.1223	-0.000720
1.0487	1.0228	0.0250	0.1880	0.001090
1.0077	1.0133	-0.0056	0.3006	0.001138
1.0264	0.9860	0.0401	0.3665	0.005967
1.0222	0.9892	0.0328	0.5301	0.000176
0.9995	1.0286	-0.0286	0.6144	-0.022285
0.9579	1.2246	-0.2457	0.7769	-0.021386
0.9958	1.2899	-0.2587	0.8617	-0.040975
1.0000	1.3963	-0.3338	1.0000	-0.072278

The value of the integral $I = -0.07$ appears to indicate that the data are thermodynamically consistent

C. Now we must calculate the coefficients τ_{12}, G_{12}, τ_{21} and G_{21} for the NRTL model. Using the values of the liquid activity coefficients at infinite dilution along with the formulas we developed in Part A. of this problem we obtain

$$\ln \gamma_1^{\infty} = \tau_{21} + \tau_{12} \exp(-0.3\tau_{12}) = 0.078$$

$$\ln \gamma_2^{\infty} = \tau_{12} + \tau_{21} \exp(-0.3\tau_{21}) = 0.334$$

Solving these two equations with the Excel Solver we obtain

$$\tau_{12} = 1.4450 \Rightarrow G_{12} = 4.242$$
$$\tau_{21} = -0.8587 \Rightarrow G_{21} = 0.4237$$

and since
$$\ln G_{12} = A_{12} + B_{12}/T$$
$$\ln G_{21} = A_{21} + B_{21}/T$$

we may set
$$A_{12} = 1.445 \quad B_{12} = 0$$
$$A_{21} = -0.8587 \quad B_{21} = 0$$

We can now perform a non-linear regression using the model

$$\boxed{P = x_1 \gamma_1 P_1^{sat} + x_2 \gamma_2 P_2^{sat}}$$

where γ_1, γ_2 are estimated using the NRTL model. Numerical results and values of the constants A_{12}, A_{21} are shown

(5)

Non-Linear Regression with the NRTL model

Experimental			NRTL			
γ_1	γ_2	$\ln(\gamma_1) - \ln(\gamma_2)$	γ_1	γ_2	Predicted Pressure (kPa)	$(\text{Residual})^2$
1.0607	1.0000	0.0589	1.0607	1.0000	9.4500	0.0000
1.0555	0.9676	0.0869	1.0523	1.0001	11.9177	0.0828
1.0303	0.9949	0.0350	1.0450	1.0005	14.3071	0.0161
0.9625	1.0087	-0.0469	1.0355	1.0015	17.8529	0.3757
1.0487	1.0228	0.0250	1.0259	1.0032	22.2363	0.2244
1.0077	1.0133	-0.0056	1.0141	1.0069	29.6343	0.0109
1.0264	0.9860	0.0401	1.0094	1.0092	33.9290	0.1095
1.0222	0.9892	0.0328	1.0026	1.0146	44.5691	0.4501
0.9995	1.0286	-0.0286	1.0010	1.0168	50.0648	0.0006
0.9579	1.2246	-0.2457	0.9999	1.0191	60.7058	4.1039
0.9958	1.2899	-0.2587	0.9999	1.0192	66.2797	0.0081
1.0000	1.0172	-0.0171	1.0000	1.0172	75.3800	0.0000

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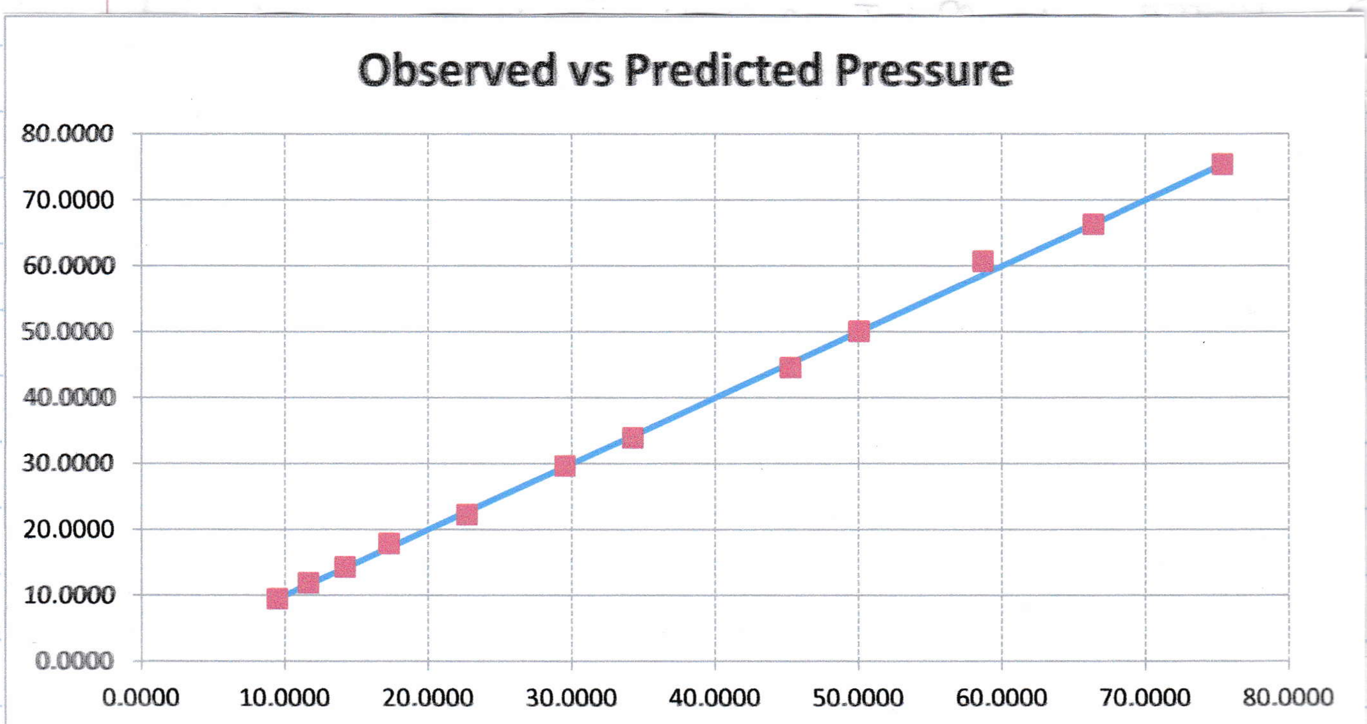
SSQ	5.33227
A12	-0.58924
A21	0.76206
B12	0.00000
B21	0.00000
C	0.30000

Parity Plot of Predicted Pressure vs. Observed



Notice that the initial guesses for A_{12} & A_{21} are very bad because the calculation of γ_1^∞ and γ_2^∞ was not very good.